INTERNAL · PRODUCTION REFERENCE

Machine Element Brochure

Every machine on our workshop floor and in our site kit — what it is, and what we use it for.

45 MACHINES · 8 SECTIONS

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HOW TO READ A CARD

- what it is — the machine, in one line
- purpose — where and why we use it in production

This document carries no cost tiers; machine time is quoted against current production rates.

I Carpentry

CNC Router

A computer-controlled 4-axis routing machine that follows programmed toolpaths to cut, carve, drill and profile sheet materials such as MDF, plywood, acrylic, foam and composites.

Used to produce accurate wall panels, counters, decorative forms, signage components and repeatable stand parts with clean, consistent dimensions.

Sliding Table Saw

A precision panel saw with a sliding carriage that supports boards during straight, angled and repeat cuts.

Used to size MDF, plywood, laminates and timber panels accurately for exhibition walls, flooring, counters, furniture and joinery.

Sliding Compound Mitre Saw

A stationary saw with a sliding arm and tilting head that produces accurate crosscuts, mitres and compound angles in timber and board sections.

Used to cut framing timber, trims, skirtings and mouldings to repeatable lengths and angles for stand structures, platforms and joinery.

Handheld Circular Saw

A portable power saw with a rotating toothed blade and adjustable depth and bevel settings for straight cuts in board and timber.

Used to break down sheet materials, trim panels and make straight corrective cuts in the workshop and during on-site build-up.

Jigsaw

- A handheld power saw with a narrow reciprocating blade for curved, irregular and internal cuts in wood, MDF, acrylic and thin sheet materials.
- Used for openings, notches, decorative shapes and quick adjustments during workshop fabrication and on-site stand installation.

Spindle Moulder / Shaper

- A stationary woodworking machine that uses rotating profile cutters to shape edges, grooves, rebates and mouldings in timber or board materials.
- Used to produce custom trims, decorative profiles, joinery details and finished edges for counters, wall features and exhibition furniture.

Thickness Planer

- A woodworking machine that feeds timber through a rotating cutter head to produce a uniform thickness and smooth, parallel faces.
- Used to calibrate timber before fabricating frames, counters, decorative structures and custom exhibition joinery.

Electric Hand Planer

- A handheld power planer with a rotating blade drum that removes thin, even shavings from timber faces and edges.
- Used to trim doors and panels, level frames, ease edges and correct fit during assembly and site installation.

Trimmer

- A compact handheld router fitted with small cutters for edge trimming, laminate overhang removal and light profiling.
- Used to flush-trim laminate and edge banding, round over edges and finish visible details on counters, cabinets and stand joinery.

Edge Banding Machine

- A machine that applies, bonds and trims PVC, ABS, veneer or laminate edging to exposed board edges.
- Used to conceal raw MDF or plywood edges and create clean, durable finishes on counters, cabinets, shelves and stand furniture.

Hand Power Drill

- A handheld drilling tool used to create holes in wood, MDF, acrylic, aluminium and light-gauge metal sections.
- Used for pilot holes, anchor holes and general assembly works during stand fabrication and site installation.

Compact Driver

- A compact cordless driver or impact driver designed for driving screws, self-tapping fasteners and fixing hardware efficiently.
- Speeds up furniture assembly, wall framing, panel fixing and snagging in tight or hard-to-reach areas.

II Metal Fabrication

Abrasive Chop Saw

ⓘ A metal-cutting machine that uses a high-speed abrasive disc to cut through steel and aluminium sections, leaving a hot edge and light burr.

➔ Used to cut steel and aluminium tubes, channels and sections to length for stand frames, supports, signage and fabricated metal features; cut ends are dressed before welding.

Bench Grinder & Metal Polisher

ⓘ A fixed bench machine fitted with grinding, wire or polishing wheels for metal preparation and finishing.

➔ Used to deburr cut edges, sharpen tools, smooth welds, prepare fabricated parts and polish visible metal components.

Angle Grinder

ⓘ A handheld rotary grinder fitted with cutting, grinding or flap discs for metalworking and surface preparation.

➔ Used to cut metal sections, trim bolts, smooth welds, deburr edges and prepare fabricated components for finishing.

Buffing & Polishing Machine

ⓘ A motor-driven finishing machine that uses buffing pads, polishing discs and compounds to refine acrylic, metal and coated surfaces.

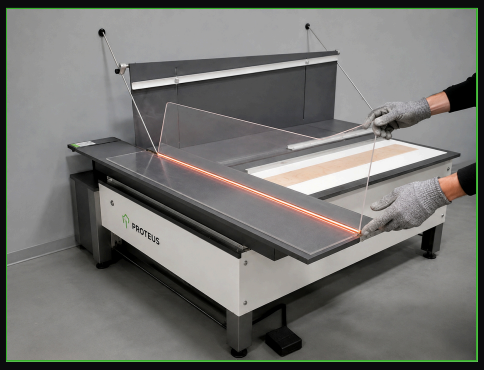
➔ Used to remove fine scratches, smooth edges and achieve a clean gloss finish on visible stand and signage components.

III Acrylic

CO2 Laser Cutting & Engraving Machine

A computer-controlled laser machine that uses a CO2 beam to cut or engrave acrylic, wood, paper, fabric, leather and selected plastics.

Used to produce acrylic letters, logos, signage, templates, display parts and intricate decorative elements with fine detail.



Acrylic Line Bending Machine

A heating machine that softens a narrow line in acrylic or thermoplastic sheet so it can be bent accurately along a defined axis.

Used to form clean angles for display holders, shelves, plinths, signage, covers and custom exhibition components.

Letter Bending Machine

A specialized machine that bends aluminium or stainless-steel coil into accurate channel-letter shapes for signage production.

Produces 3D brand letters and illuminated logo returns used on fascia signage, reception features and exhibition graphics.

Plunge Router

- A heavy-duty plunge router that lowers the cutter vertically into the workpiece for controlled routing operations.
- Used to cut grooves, rebates, recesses, hinge pockets, edge details and precise openings in carpentry and acrylic components.

Heat Gun

- A hot-air tool that delivers controlled heat through adjustable temperature settings and nozzle attachments.
- Used for vinyl application, edge-band softening, acrylic forming support, heat-shrinking, adhesive activation and surface drying.

IV Printing & Graphic

Eco-Solvent Printer

- A 1.2m roll printer that uses eco-solvent inks to produce durable graphics on vinyl and sticker media.
- Used for wall branding, decals, signage and exhibition graphics requiring strong colour and indoor or outdoor durability.

UV Roll-to-Roll Printer

- A 3.2m large-format printer that applies and instantly cures UV ink onto flexible roll media such as vinyl, banner, fabric and backlit film.
- Used to produce durable, high-quality exhibition graphics with fast drying, strong colour and compatibility across multiple media types.

Vinyl Plotter / Cutting Machine

- A computer-controlled blade cutter that follows vector artwork to cut adhesive vinyl, masking film and printed graphic media.
- Used to produce lettering, logos, decals, wayfinding, application masks and contour-cut branding for exhibition stands.

Cold Laminating Machine

- A pressure-roll machine that applies protective films, mounting adhesives or textured overlays to printed graphics without heat.
- Used to protect prints, add the required finish, mount graphics and improve durability during transport, installation and exhibition use.

v Paint

Spray-Paint Booth

An enclosed and ventilated finishing area with filtered airflow and extraction for controlled application of paint, primer and coatings.

Used to achieve cleaner, safer and more consistent finishes while containing overspray, dust and fumes during stand production.

HVLP Spray System

A high-volume, low-pressure spray system designed to apply coatings with controlled atomization and reduced overspray.

Used for fine finishing of joinery, furniture, counters and detailed stand features where surface quality and material control are important.

Orbital Sanding Machine

A power sander whose pad moves in an orbital or random-orbit pattern to smooth surfaces with reduced visible sanding marks.

Used to prepare MDF, timber, filler and painted surfaces before priming, coating or applying the final finish.

Drywall / Wall & Ceiling Sander

- A long-reach powered sander with a rotating head and integrated dust extraction for sanding large wall and ceiling surfaces.
- Used to level filler and skim coats on stand walls, bulkheads and ceilings before priming and painting, with dust captured at the head.

Spray Gun

- A handheld paint spray applicator used with compressed air to atomize paint, primer or lacquer into a fine finish.
- Provides smooth, even coating on stand walls, joinery, furniture, feature elements and touch-up works.

VI Assembly & Support

Air Compressor

A motor-driven unit that compresses and stores air for use by pneumatic tools, spray equipment and workshop machinery.

Used to power spray guns, nailers and air tools, support cleaning operations and supply equipment during fabrication.

Compressor Nail Gun

A pneumatic nailing tool powered by an air compressor for rapid and repetitive fastening of timber components.

Speeds up carpentry assembly, backing-frame fabrication, trim fixing and panel-support construction in the workshop.

Dust Extraction System

A workshop extraction system that captures sawdust and airborne particles at the machine source through ducts, filters and collection bags.

Used to maintain a cleaner and safer production area, protect machinery and improve visibility and finish quality during cutting and sanding.

Industrial Vacuum System

- A heavy-duty, high-capacity vacuum designed for continuous removal of fine dust, debris and production waste.
- Used to clean machinery, floors and fabrication areas and support dust control throughout stand production.

Electric Air Blower

- An electric blower that delivers a concentrated, adjustable stream of air through a nozzle attachment.
- Used to clear dust and debris from panels, frames and floors before painting, graphics application and client handover.

Portable Generator

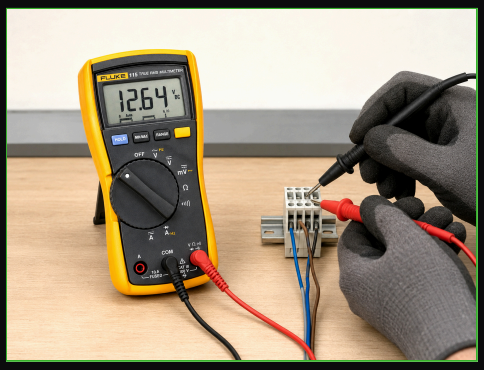
- A fuel-powered generator unit that produces temporary electrical supply independent of mains power.
- Provides temporary electrical power during installation, outdoor work or locations without stable supply.

VII Electrical

LED Soldering Station

A temperature-controlled soldering unit with an iron, stand and interchangeable tips for precise low-voltage electrical work.

Used to assemble, connect and repair LED strips, modules, connectors and wiring for illuminated signage and exhibition stand features.



Multimeter

A digital electrical testing instrument used to measure voltage, continuity, resistance and related circuit values.

Helps technicians test power lines, LED circuits, adapters, sockets and faults during stand assembly and troubleshooting.

VIII Material Handling & Installation

Forklift

- A powered industrial truck with a lifting mast and forks for raising and transporting heavy or palletized loads.
- Used to move and load sheet materials, crates, machinery and fabricated stand components around the workshop and loading area.

Manual Pallet Lift

- A hand-operated pallet truck with forks and a hydraulic lift mechanism for moving palletized materials over short distances.
- Used to transport boards, crates, hardware and packed stand components safely within the workshop and loading areas.

Manual Duct Lifter

- A manual material lift with a winch-operated mast designed to raise lightweight overhead items to working height, typically up to around 6 metres.
- Assists in lifting signage, light trusses, fascia frames, ducts and suspended elements safely during build-up and installation.

Scaffolding

- A modular access system or mobile scaffold tower that provides temporary elevated working platforms. Goes up to heights of 6m.
- Allows safe working at height for painting, branding application, electrical fixing, rigging and finishing of exhibition stands.

Vacuum Glass Lifter

- A lifting device that uses vacuum suction cups and a pump system to securely grip smooth sheet materials.
- Used to lift, position and install glass, acrylic and smooth panels more safely while reducing manual handling and the risk of damage.

Storage Trolley

- A flat-deck platform trolley on four castors with a folding push handle, sized for moving stock and finished parts around the floor.
- Used to shift materials, hardware and completed elements between workstations and out to loading, with the handle folding flat for storage.

Flatbed Platform Trolley

- A wheeled trolley with a low flat load deck and push handle for moving boxed, crated and loose materials horizontally.
- Used to move tools, hardware, crates and packed stand components across the workshop, loading bay and exhibition floor.